Is the Jacobian Space a Global Workspace?

David J. Chalmers

In "Verbalizable Representations form a Global Workspace in Language Models", Gurnee et al define a special component of the activation space in a language model: the Jacobian space, or J-space for short.[1] Representations in the J-space are *verbalizable* in that they are especially likely to influence the words used in the language model's outputs. Gurnee et al argue that the J-space functions as a mechanism for conscious access (or access consciousness). They also argue that it closely resembles a well-known proposed mechanism for conscious access: the global workspace, put forward by Bernard Baars (1988) and developed further by Stanislas Dehaene (2014) and colleagues.

These claims are exciting for a number of reasons. The global workspace theory is perhaps the most popular scientific theory of consciousness, and finding a global workspace in a language model suggests that we should take the question of consciousness in these models seriously. Gurnee et al disclaim any implications for subjective experience, or phenomenal consciousness (the kind of consciousness that generates the hard problem), but many others see access consciousness and phenomenal consciousness as closely connected. Furthermore, even implications for access consciousness and for the associated distinction between conscious and unconscious states (understood in terms of access) are extremely interesting.

I will argue that while the J-space is interesting and powerful, its connections to the global workspace and to access consciousness have been somewhat overstated. Where the global workspace is concerned, much of the putative evidence that the J-space is "workspace-like" can be better understood as evidence that the J-space is involved in access consciousness. Where access consciousness is concerned, I will argue that verbalizability is only weakly tied to access consciousness in the usual sense, so that the case for robust access consciousness is weaker than it could be. I'll also discuss the upshot for phenomenal consciousness in language models.

[1]Thanks to Jack Lindsey for extensive discussions, and to an Eleos discussion group and an audience at the Computational Consciousness Science workshop at the NYU Cognitive Computational Neuroscience conference.

Preliminaries

The J-lens and the J-space are (very roughly) ways of associating activation vectors in a language model with a set of output tokens that the vector is especially apt to influence.

The Jacobian lens can be seen as a refinement of a familiar interpretability device, the *logit lens* (Nostalgebraist 2020). The logit lens relies on the standard unembedding matrix, which is standardly used at the output stage of any language model, associating the activations in the final layer of a transformer with a score for each token that can be converted into a probability of outputting that token. The logit lens simply applies the same unembedding matrix to earlier layers of a transformer, associating any activation in that layer with a score for each token. For example, if a model is asked "What is 2+3?", the token "5" starts to receive high scores in the logit lens for intermediate layers and receives a very high score in later layers.

The J-lens does something similar to the logit lens but with some improvements. The improvements will not play a central role in my discussion, and indeed Gurnee et al note that the logit lens often performs reasonably well on their key tasks, although not as well as the J-lens. The key improvements are that the J-lens applies a mathematical Jacobian to compensate for transformations in co-ordinate systems between intermediate layers and the final layer, and that the J-lens makes extensive use of averaging in order to compute an average score for an activation vector across many different contexts. The first change helps to improve the comprehensibility of the results in intermediate layers, while the second change helps to assess the tokens that are reliably associated with a given activation vector rather than those that are associated on a given occasion.

The J-space uses the J-lens to define a special subframe of activation space consisting of vectors that correspond to single tokens (vectors that yield the highest score for those tokens), along with weighted combinations of a small number of these token-specific vectors. We can then associate any given activation vector with a vector in J-space which is closest by a certain definition to the original vector. Any given activation vector can be written as the sum of a J-space component with a residue which is a non-J-space component. The J-space component can be seen as encoding a small set of tokens for which that vector has especially strong influence. Given an activation vector in J-space, the output is especially likely to involve the corresponding tokens. That is, this vector is likely to yield *verbalizations* of the relevant tokens.

Gurnee et al suggest that this property of the J-space is especially reminiscent of *access consciousness*, or conscious access. Access consciousness is usually defined as availability for report, reasoning, and action. In somewhat more detail, a mental state is access conscious if it is available for verbal report and for flexible (as opposed to rigid) and deliberate (as opposed to reflexive or automatic) reasoning and action. The rough idea is that because J-space vectors are distinctively verbalizable, they are available for report, and they also turn out to be available for reasoning and action.

Access consciousness is typically contrasted with *phenomenal consciousness*, or subjective experience. A system is phenomenally conscious if there is something it is like to be that system. Where access consciousness is defined in functional terms and is relatively unmysterious, phenomenal consciousness is defined in experiential terms and is often thought to be especially difficult to explain. The main focus of this article is access consciousness, not phenomenal consciousness.

Gurnee et al characterize access consciousness by outlining “several functional properties that are commonly held to distinguish consciously accessible information from unconscious processing”. These include that the information is reportable, subject to top-down control, is the medium of deliberate reasoning, permits flexible generalization, and is selective. These five criteria correspond at least roughly to the standard criteria: reportability and deliberate reasoning correspond directly, while top-down control corresponds roughly to deliberate action, and flexible generalization corresponds roughly to flexible reasoning and action. Selectivity is not usually understood as part of the definition of access consciousness, but it is certainly part of the “folk psychology” of access consciousness that only a limited amount of information is conscious at a time.

The *global workspace theory* is a theory which grounds access consciousness in certain computational features. (The well-known *global neuronal workspace theory* also brings in neurobiological features, but these features are less relevant for the AI case.) Roughly, a representation is consciously accessible when it is posted to a shared “global workspace”. The global workspace is a processing hub that integrates information from many specialized unconscious modules, allowing that information to be used for flexible internal reasoning and report. The workspace is limited in capacity, so entry is competitive and subject to attentional modulation. A vector enters the workspace when its activity is amplified via recurrent feedback in a manner known as “ignition”. Upon ignition, the contents of the workspace are broadcast to the unconscious modules.

The global workspace is typically understood as a mechanism for conscious access, but its defining features go well beyond the defining features of access consciousness. Access consciousness is defined functionally in psychological terms, whereas the global workspace is defined in largely computational terms. There are many competing computational theories of access consciousness, and the global workspace theory is just one. It seems possible in principle to have access consciousness without the key features of a global workspace: a centralized processing hub, specialized modules, recurrent feedback, ignition, and more. But global workspace theorists argue that at least in humans, access consciousness is realized by a global workspace.

Access consciousness and the global workspace[2]

Gurnee et al also define a notion of "workspace-like" representations. They "define a subset of vector representations as workspace-like if it satisfies the following properties, which mirror the properties characteristic of conscious access described above". The five properties are verbal report, directed modulation, internal reasoning, flexible generalization, and selectivity, each now characterized in computational terms. Surprisingly given the name "workspace-like", these five properties reflect the generic characterization of conscious access and not any of the further distinctive claims of global workspace theory.

This enables us to make an initial objection to Gurnee et al's claims:

1. *The "workspace-like" properties of J-space are properties of conscious access, not distinctive properties of the global workspace theory.*

Gurnee et al are quite explicit that workspace-like representations are defined in terms of properties mirroring the properties of conscious access. Because of this, the label is misleading. A natural correction would be to replace the term "workspace-like" with "conscious-access-like", or just "access-like" for short. If we did that consistently, the conclusion of the core arguments of the paper would be that the J-space is access-like, not that it is workspace-like.[3]

Gurnee et al go on to say "we do not claim that language models reproduce the full architecture global workspace theory ascribes to the brain", which include "specialized, encapsulated processors competing for entry to a workspace that broadcasts back to them through recurrent connections" along with "the sharp, competitive 'ignition' that characterizes workspace entry in the brain. Our findings suggest that the J-space achieves many of the functional properties of the global workspace in the brain, while sharing only some of its architectural properties."

Since these architectural properties are the key distinctive properties of the global workspace, over and above the functional properties of access consciousness, this already makes clear that the evidence that the J-space is a global workspace (over and above evidence that it supports access consciousness) is weaker than the title suggests.

[2]My conclusions in this section (especially objections 1 and 2) are closely related to those drawn by Butlin et al (2026) in their commentary on Gurnee et al, although my reasons are somewhat complementary to theirs.

The point can be clarified by reviewing the key computational and architectural properties of a global workspace (over and above the properties of conscious access). These include that the global workspace is (1) a system that (2) broadcasts information to (3) unconscious specialized modules via (4) recurrent loops while also (5) integrating information from these modules with (6) selective entry and (7) ignition on entry.

In what follows I will review how well the J-space has these properties, with special attention to the first two.

2. *The J-space is not a (causal, dedicated, privileged, unified) system.*

The classical global workspace is understood as a *system* that contains selected representations that are broadcast and integrated. This system is an architectural component of the cognitive system as a whole, one that is dedicated to broadcast and integration, and one that serves as the primary mechanism for broadcast and integration in the cognitive system. Baars's classical workspace is an "information exchange" modeled on the "blackboard" architecture of the Hearsay model (Erman et al 1980) where a single blackboard system serves as the locus of broadcast and integration. Dehaene and Naccache's global neuronal workspace posits a distinctive neuronal population: a subset of cortical pyramidal cells with long-range excitatory axons, dense in prefrontal, cingulate and parietal cortex, with "tight bidirectional connectivity" unifying the neurons in the workspace. This dedicated population is responsible for broadcast and integration.

We might call this a *substantial workspace*. It is a causal system (one that plays a clear causal role), a dedicated system (a primary mechanism or population that is dedicated to broadcast and integration), a privileged system (a system unlike other systems in that it is devoted to broadcast), and a unified system (one integrated system rather than many fragments).[4]

The substantial workspace contrasts with what we might call a *minimal workspace*. A minimal workspace is simply a collection of representations that are selectively broadcast. There is no requirement that these representations form a single system or that there is a single mechanism of broadcast. There could be a fragmented workspace involving multiple systems, or there could be broadcast mechanisms with no dedicated systems at all.[5]

[4]Butlin et al 2026 likewise speak of the global workspace as requiring an integrated "unified stream" and suggest that evidence so far does not establish such a stream in language models.

[5]Do Dehaene and colleagues endorse a substantial workspace or a minimal workspace? The global neuronal workspace they describe certainly seems substantial, involving a dedicated, privileged, and unified system that plays a clear causal role in broadcast and integration. There are occasional remarks suggesting a more minimal attitude: for

Daniel Dennett (2001) argues for the minimal workspace model over the substantial workspace model. He holds that global broadcast ("cerebral celebrity" or "fame in the brain") is crucial to consciousness, but that the mechanisms of broadcast may involve multiple fragmented systems (as in his multiple-drafts theory). On this model, all it takes to have a workspace is that some representations are broadcast while others are not. Since being broadcast is roughly equivalent to being accessed, the minimal workspace model is roughly equivalent to a model where some representations are access-conscious and some are not.

One way to capture the distinction between these two models is as follows: on the substantial workspace model, representations are broadcast (or accessed) because they are in the workspace. On the minimal workspace model, representations are in the workspace because they are broadcast.

Is the J-space more like a substantial workspace or a minimal workspace? At least on a first look, it appears more like a minimal workspace. The J-space is defined by a number of key activation vectors. These vectors are selected precisely because these are the ones that have a strong disposition to influence output tokens. We might say: they are in the J-space because they are accessible, or disposed to be broadcast. This is the mark of a minimal workspace.

It might be objected that the J-space is a dedicated subspace of activation space, and that a subspace can be considered a substantial workspace. Now, a subspace is very far from a traditional component of cognitive architecture. Instead of some representations being in the subspace, every representation will be decomposable with a component in the subspace and a component outside. Furthermore, the J-space is not even a subspace, but instead a union of many k-dimensional cones, one for every set of k J-lens vectors. It is far from clear that this gerrymandered object is a genuine component of cognitive architecture, playing a causal role in the workings of the model and not just an epistemological role in interpretability.

Potential evidence that J-space plays a genuine causal role comes from Gurnee et al's extensive experimental evidence that vectors in the J-space function quite differently from vectors outside the J-space. However, it is far from clear that this makes J-space a key causal mechanism. For example, the logit lens shows many of these effects, if not as robustly as the J-lens. There are likely

example "It is the style of activation (dynamic long-distance mobilization), rather than its cerebral localization, which characterizes consciousness" (Dehaene and Naccache 2001). My reading is that Dehaene and colleagues endorse a minimal account of conscious access (conscious access requires only broadcast) but a substantial account of the global workspace (global workspace involves a substantial system for broadcast), along with the key claim that the global workspace realizes broadcast (and therefore conscious access) in humans.

many lenses and spaces that show even more robust effects along these lines. For example, a recent study (Blank et al 2026) has shown that a related R-lens works better than a J-lens in analyzing earlier layers. It is very likely that the various striking features of the J-space (for example, the way the system degrades when J-space vectors are ablated) apply equally to the R-space. There may well be many other spaces of which this is true. Merely demonstrating that the J-space has these properties does not establish that it is a privileged mechanism. In the absence of stronger evidence, it seems likely that the J-lens is a useful tool for interpretability without corresponding to a privileged system or mechanism.[6]

Overall, perhaps we can say that the J-space lies somewhere on a spectrum between a minimal workspace and a substantial workspace, but that evidence that it supports a substantial workspace is currently weak.

3. *The J-space supports only a limited form of broadcast.*

A broadcast system is one for which a single sender sends a signal to many receivers. In a classical global workspace system, the workspace sends a signal to many parts of the system distributed widely in the brain.

The J-space does not seem to broadcast information in this sense. The direct influence of the J-space in a given layer is restricted to the following layer. The indirect influence extends to later layers, while perhaps the most crucial influence (measured by the Jacobian) is on the final layer and the associated outputs. Other parts of the system (such as earlier layers and associated mechanisms) are not influenced by this J-space at all, except indirectly by recirculating outputs as inputs. This is very different from classic workspace's simultaneous parallel influence on multiple modules.

Gurnee et al (4.3) argue that the J-space has an unusually strong influence on the crucial final

[6] Many of the results presented by Gurnee et al to establish that the J-space is special involve showing that vectors in the J-space are superior on various metrics to vectors outside the J-space. Results of this sort on their own cannot establish that J-space is special, since these results are consistent with there being many K-spaces such that vectors in the K-space are just as superior to vectors outside the K-space. We would need results showing that vectors in the J-space are superior overall to vectors in the K-space. Gurnee et al do occasionally present results of the latter sort, especially in 4.3 where they identify broadcast heads for six different populations of vectors and show that broadcasts heads corresponding to J-lens vectors score higher on various metrics than heads corresponding to the other five populations. Of course merely beating five other populations is not enough to establish that J-space is privileged here. Evidence suggesting that they beat all other populations would count for much more, but no evidence of that form is presented in the paper as far as I can tell.

layer and on associated outputs. But this seems best characterized not as broadcast in the classical sense (direct transmission from sender to receiver) but as *influence*. The J-space exerts key indirect influence on the final layer and on the associated outputs. Indeed, the J-space is largely defined by this sort of influence. It serves as a model of how vectors in hidden layers influence outputs.

Likewise, there is perhaps something broadcast-like about the way that J-space activity influences non-J-space activity in subsequent layers. Gurnee et al (4.3) show evidence of attention heads that are especially sensitive to J-space vectors. Of course strictly speaking, every subspace influences activity outside that subspace. The J-space just has somewhat larger influence than some other subspaces. Again, this is more accurately characterized in terms of influence rather than broadcast.

As a model of consciousness, this resembles Dennett's "fame in the brain" model where consciousness is understood in terms of influence. It is perhaps even closer to an earlier model of Dennett's (1978) where consciousness is grounded in effects on a speech module. Here it is one sort of influence that matters most: influence on linguistic outputs. This is also the sort of influence that matters most for vectors in the J-space.

Models that analyze consciousness in terms of influence on outputs are an important class of models. But prima facie, they are distinct from global workspace models that analyze consciousness in terms of global broadcast to all components of a cognitive systems. The J-space does not seem to engage in broadcast in this sense.

4. *The J-space lacks key computational features of the global workspace including modules, recurrent processing, and others.*

Recall that our key properties of a global workspace are: (1) a system that (2) broadcasts information to (3) unconscious specialized modules via (4) recurrent loops while also (5) integrating information from these modules with (6) selective entry and (7) ignition on entry. So far I have discussed (1) and (2). I will here briefly review (3) to (7).

Regarding modules: Gurnee et al say "We do not provide evidence that non-J-space processing consists of encapsulated modules".

Regarding recurrence: They say "the form of broadcast we identify takes place not via recurrent connections, but rather over the course of the model's depth". The architecture of a standard transformer is almost entirely feedforward, with little room for recurrent feedback. Gurnee et al's suggestion about depth is an alternative to the standard global workspace architecture and not a version of it.

Regarding integration: the issues here are analogous to those in the case of broadcast. It is reasonable to hold that the J-space is integrating information in some sense, though it may not be a privileged mechanism for integration.

Regarding capacity limits: The J-space as designed involves sparse combinations of 25 or so J-lens vectors, so it has a built-in limited capacity in storing these vectors. It therefore has a limited capacity in representing the corresponding tokens, although a somewhat higher capacity than usually supposed for a human global workspace.

Regarding ignition: Gurnee et al argue that there are "ignition-like effects" in J-space over intermediate layers of a language model as a representation becomes dominant, with a relatively quick transition from a continuous distribution to a bimodal distribution. Still, this degreed model of presence in the workspace differs from the all-or-nothing status of standard ignition models, leading Dehaene and Naccache (2026) to say that "Ignition remains to be fully demonstrated" in the J-space.

Overall, the J-space shows at best limited versions of integration, capacity limits, and ignition, and it does not seem to involve recurrence or modules at all. So the J-space seems to be missing a number of key computational features of a classic global workspace.

Stepping back, proponents of J-space as a global workspace could respond to all of these objections by saying that J-space at least supports a minimal workspace in the sense of having the five main "workspace-like" (i.e. access-like) features that they argue for, including reportability, top-down control, deliberate reasoning, flexible generalization, and selectivity. For now, I am happy to concede the case for a minimal workspace along these lines (though I will shortly raise doubts about reportability). But a version of GWT that invokes a minimal workspace is less an empirical computational theory than a version of the pretheoretical psychology of conscious access. And the thesis that J-space is a workspace now threatens to collapse into the claim that it supports access consciousness. The case that it is a distinctive global workspace that underlies access consciousness disappears.

Some proponents of LLM consciousness will be happy with the conclusion that there is a case that the J-space supports access consciousness, whether or not it supports a global workspace. After all, access consciousness is a form of consciousness, even if it is not phenomenal consciousness. And many theorists are interested in the global workspace precisely because it is a potential mechanism for access consciousness. But in the next section I will argue that even the evidence for access consciousness in these models can be questioned.

Verbalizability and reportability

The key to using the J-space as a model of conscious access is that J-space vectors are *verbalizable*, in the sense that they have distinctive effects on verbal outputs. Gurnee et al suggests that this mirrors a key aspect of access consciousness—that access-conscious representations are *reportable*, in that subjects can verbally report the contents of these representations. My doubts here concern whether there is really a tight link between verbalizability and access consciousness, or between verbalizability and reportability.

5. *Many contents of access consciousness are not verbalizable.*

It is a truism that many contents of consciousness are hard to put into words. This holds for spiritual and psychedelic experiences, but it equally applies to mundane sensory experiences, such as novel tastes and shapes, and to some sorts of conscious thought, such as "unsymbolized thinking" (Hurlburt and Akhter 2008). All of these experiences are not just phenomenally conscious but access conscious: people can report on having them and reason about them. But subjects often cannot put them into words perhaps beyond "There's that smell again".

Gurnee et al acknowledge that human access consciousness includes nonverbalized aspects, but suggest that pure language models may be different, because their only inputs and outputs are verbal tokens, whereas humans have sensory inputs and bodily actions as outputs. As a result of this restriction, it is natural for access consciousness in pure language models to include only verbal or verbalizable representations. This restriction applies only to pure language models, and not to multimodal models, significantly restricting the scope of the overall thesis. Even where pure language models are concerned, it is plausible (evidence?) that language models can develop new representations that do not correspond to tokens in their training data. Even if they are definable in terms of existing tokens, then like other concepts that require multiple tokens (e.g. *blackmail*), these will not show up in the J-lens or the J-space.[7]

[7] Two further worries about access consciousness without verbalizability are specific to Gurnee et al's specific and somewhat idiosyncratic notion of verbalizability, which we might call J-verbalizability. One point is that J-verbalizability (and therefore the J-space) captures accessibility averaged across contexts, but conscious access is typically tied to a specific context. A related point is that access consciousness is a trigger-relative dispositional notion (a representation is access conscious if it would be accessed were certain triggers to obtain), while J-verbalizability is not trigger-relative in this way (because it is an average across all triggers or all contexts). A third point is that concepts expressed by multiple tokens (such as *blackmail*) will not be J-verbalizable even if they are access conscious. As a result of all three of these points, many representations will be access-conscious but not J-verbalizable.

6. *What matters for the J-space may not be effects on verbal outputs, but effects on outputs in general.*

Another potential objection is triggered by Gurnee et al's observation above that most outputs from a pure language model are verbal tokens (corresponding to words or parts of words). This means that in a pure language model, verbalizability (influence on verbal outputs) roughly coincides with what we might call outputability (influence on outputs more generally). And it raises the possibility that the distinctive properties of the J-space are connected less to verbal outputs than to outputs in general.

This would be an easy matter to test. We need only run J-space experiments in a multimodal model with non-verbal and non-textual outputs as well as verbal outputs. GPT-4o is a paradigmatic example, including image and audio outputs as well as textual outputs. Other relevant multimodal models include video models and robotic action models. For all of these models, we can easily define a J-lens and a J-space in terms of influence on multimodal outputs, not just textual outputs. The empirical question is whether the various J-space effects will arise on these multimodal models too.

My prediction is that we will find robust J-space effects in multimodal models, not just pure language models, and probably even in relatively unimodal moswla with only image outputs or only audio outputs, and so on. If so, it looks like J-space effects do not arise so much from the verbalizability of relevant vectors as from their outputability, or their distinct effects on outputs of all sorts.

Gurnee et al might simply accept this consequence and accept that the effects generalize from verbal to nonverbal outputs. But this will greatly change the conclusions of the paper. The title "Verbalizable Representations Yield a Global Workspace in Language Models" will be better rephrased as "Output-Influencing Representations Yield a Global Workspace in Language Models". And much of their discussion is motivated by analogies with the human case where verbalizable or reportable representations are supposed to play a special role, distinct from the role played by merely output-influencing representations. If my prediction is correct, it looks like as if multimodal LLMs behave quite different from humans here, with the distinctive effects arising from outputability rather than reportability.

A more general worry here is that Gurnee et al's framework may not deliver a good model of the unconscious: even J-unverbalizable representations often look more like access-conscious representations (because they can impact report, reasoning, and action at least in some crucial circumstances) than like traditional access-unconscious representations.

Similarly, many theories of consciousness, including the global workspace theory, tie consciousness (sometimes both access consciousness and phenomenal consciousness) especially to reportable representations rather than output-influencing representations. To a first approximation, reportable representations are conscious while mere output-influencing representations are often unconscious. If Gurnee et al's J-space effects arise from outputability rather than from reportability or verbalizability, there is much less of a connection between J-space and these theories of human consciousness than one might have thought.

7. *Many verbalizable representations are not access conscious.*

The dissociation between verbalizability and access consciousness goes both ways, in that unconscious processing may involve verbalizable elements. Consider a classic Freudian case in which a subject has unconscious thoughts about their mother that lead them to frequently say the word "mother". These thoughts will be verbalizable in Gurnee et al's sense (they systematically influence verbal outputs) and their counterparts in language models will show up in the J-space, but they will not be access conscious. As a result they will not be part of the global workspace if there is one.

8. *Verbalizability is not the same as reportability.*

Here is a potential diagnosis. Gurnee et al emphasize that access consciousness involves *reportability* as a key criterion, giving a number of different informal glosses on this notion. Before long, however, they switch from reportability to *verbalizability* as if the two notions are interchangeable. On a standard understanding, however, they are not interchangeable. In the Freudian case just mentioned, a patient's thoughts about his mother are *verbalizable*—they can bring about verbal expressions involving the word "mother"—but they are not *reportable*, in that the patient is unable to report thinking about their mother. Likewise, for ineffable experiences, one would typically say that they are not verbalizable (one cannot capture their contents in existing words), but they are still reportable, in that the subject can say that they are having an ineffable experience (e.g. "that smell again").

The key difference is that reportability requires some potential metacognitive awareness of a mental state, whereas verbalizability requires only verbal expression of the mental state without requiring awareness of it. The J-space involves a version of verbalizability but it is much less clear that it involves genuine reportability.

Of course the terms—“reportable”, “verbalizable”, “access consciousness”—can be defined in different ways and I do not mean to be insisting on my definitions. Someone could fairly distinguish *first-order reports* such as ‘There is a red square there’ from *higher-order reports* (or *subjective reports*) such as ‘I see a red square there’. They could then say that there is a reasonable sense of “reportable” that involves first-order reports and not higher-order reports. Likewise, an opponent could reasonably distinguish *first-order access consciousness* (which does not involve higher-order reports) from *higher-order access consciousness* (which includes as a criterion something like availability for higher-order reports or awareness) and say that the J-space involves the former though not the latter.

The substantive point is that there are two different phenomena here, one involving potential metacognitive awareness and one involving potential verbal expression. But without some element of metacognitive awareness, there can be no genuine subjective reports of mental states, and there can be at best a weak form of conscious access and a weak sort of global workspace. Perhaps first-order access consciousness is useful in analyzing animals without higher-order capacities. But the Freud case brings out that at least in humans who have metacognitive capacities, first-order access consciousness without higher-order access consciousness looks more like paradigmatic unconscious cognition than conscious cognition.

Just as we previously saw that a proponent of the J-space as a global workspace could always retreat to a minimal workspace, they could address the current issues by holding that a J-space may not be a higher-order workspace (one where the consuming systems include metacognitive systems), but it is at least a *first-order workspace* (one with no essential role for metacognitive awareness). But the claim that the J-space is a first-order workspace is a weaker claim involving a weaker resemblance to human consciousness.

What about phenomenal consciousness?

Gurnee et al say that their conclusions pertain only to access consciousness, not phenomenal consciousness. In their commentary, Dehaene and Naccache say that the global workspace theory is a theory of access consciousness and express skepticism about the very notion of phenomenal consciousness.

That said, there are others who regard the global workspace theory as a viable theory of phenomenal consciousness—for example, Peter Carruthers (2019) and Simon Goldstein and Cameron Kirk-Giannini (2026). Plenty of others see it as at least a potential guide to phenomenal consciousness (e.g. Butlin, Long, et al 2023). There are also many philosophers who see access consciousness as a potential guide to phenomenal consciousness (e.g., Dretske 1995; Tye 1995; Chalmers 1997; Dennett 2001; Prinz 2012). What are the consequences of our discussion for that project?

Let us first consider phenomenal GWT: a view on which GWT is a theory of phenomenal consciousness, holding that having a global workspace suffices for phenomenal consciousness. What should a proponent of phenomenal GWT say about the J-space? I've argued that the evidence that the J-space is a global workspace is weak. If so, then even if we assume phenomenal GWT, this provides weak evidence that language models are phenomenally conscious.

A proponent of J-space might respond that if GWT is a guide to phenomenal consciousness at all, this is because GWT is a guide to access consciousness, and access consciousness is a guide to phenomenal consciousness. So it remains open to a proponent of J-space to drop claims about GWT, but to say that J-space supports access consciousness and therefore phenomenal consciousness.

However, I've also argued that J-space supports at best a weak sort of access consciousness, such as first-order access consciousness. Higher-order access consciousness is a promising guide to phenomenal consciousness, not least because this involves dispositions to make subjective higher-order reports (e.g. "I am experiencing a red square"), which are arguably our best guide to phenomenal consciousness. First-order access consciousness requires only first-order reports which are less of a guide to phenomenal consciousness. If so, then even if J-space supports access consciousness, this does not seem to be the right sort to serve as a guide to phenomenal consciousness.

Still, this does bring out one potential route via which proponents of J-space could give evidence of phenomenal consciousness. They could argue that J-space supports not only first-order access consciousness but higher-order access consciousness. This would require giving evidence that language models can have some sort of introspective higher-order awareness of their internal states, where this is the sort of higher-order awareness that gives evidence of phenomenal consciousness. A related study from Anthropic (Lindsey 2025) tries to show that language models can engage in some sorts of introspection. So far it is unclear that this is the sort of introspection relevant to phenomenal consciousness, but the matter is worth investigating.[8]

To sum up: as things stand, the evidence that J-space can support phenomenal consciousness is weak, but the question remains open.

[8] I have discussed these issues in recent talks (in Bamberg, Umea, and Sydney) on "Introspection as a Guide to Consciousness in Humans and AI Systems".

References

Baars, Bernard J. 1988. A Cognitive Theory of Consciousness. Cambridge: Cambridge University Press.

Blank, Camila, Agam Bhatia, and Neel Nanda. 2026. “R-lens: Making J-lens More Faithful on Early Layers.” AI Alignment Forum, August 5, 2026. https://www.alignmentforum.org/posts/nv8oedrnLXKRzNEL9/r-lens-making-j-lens-more-faithful-on-early-layers.

Block, Ned. 1995. “On a Confusion about a Function of Consciousness.” Behavioral and Brain Sciences 18 (2): 227–247. https://doi.org/10.1017/S0140525X00038188.

Butlin, Patrick, Robert Long, Eric Elmoznino, Yoshua Bengio, Jonathan Birch, Axel Constant, George Deane, Stephen M. Fleming, Chris Frith, Xu Ji, Ryota Kanai, Colin Klein, Grace Lindsay, Matthias Michel, Liad Mudrik, Megan A. K. Peters, Eric Schwitzgebel, Jonathan Simon, and Rufin VanRullen. 2023. “Consciousness in Artificial Intelligence: Insights from the Science of Consciousness.” arXiv:2308.08708. https://doi.org/10.48550/arXiv.2308.08708.

Butlin, Patrick, Derek Shiller, Dillon Plunkett, and Robert Long. 2026. “Consciousness and Cognitive Access in LLMs: A Commentary on ‘Verbalizable Representations Form a Global Workspace in Language Models.’” In External Commentary on Verbalizable Representations Form a Global Workspace in Language Models. Eleos AI Research. https://www-cdn.anthropic.com/files/4zrzovbb/website/cc4

Carruthers, Peter. 2019. Human and Animal Minds: The Consciousness Questions Laid to Rest. Oxford: Oxford University Press. https://doi.org/10.1093/oso/9780198843702.001.0001.

Chalmers, David J. 1997. “Availability: The Cognitive Basis of Experience.” Behavioral and Brain Sciences 20 (1): 148–149. https://doi.org/10.1017/S0140525X97240057.

Dehaene, Stanislas. 2014. Consciousness and the Brain: Deciphering How the Brain Codes Our Thoughts. New York: Viking.

Dehaene, Stanislas, and Lionel Naccache. 2001. “Towards a Cognitive Neuroscience of Consciousness: Basic Evidence and a Workspace Framework.” Cognition 79 (1–2): 1–37. https://doi.org/10.1016/S0010-0277(00)00123-2.

Dehaene, Stanislas, and Lionel Naccache. 2026. “A Commentary on ‘Verbalizable Representations Form a Global Workspace in Language Models’.”

Dennett, Daniel C. 1978. “Toward a Cognitive Theory of Consciousness.” In Perception and Cognition: Issues in the Foundations of Psychology, edited by C. Wade Savage, 201–228. Minnesota Studies in the Philosophy of Science 9. Minneapolis: University of Minnesota Press.

Dennett, Daniel C. 1991. Consciousness Explained. Boston: Little, Brown.

Dennett, Daniel C. 2001. "Are We Explaining Consciousness Yet?" Cognition 79 (1–2): 221–237. https://doi.org/10.1016/S0010-0277(00)00130-X.

Dretske, Fred. 1995. *Naturalizing the Mind*. Cambridge, MA: MIT Press.

Erman, Lee D., Frederick Hayes-Roth, Victor R. Lesser, and D. Raj Reddy. 1980. "The Hearsay-II Speech-Understanding System: Integrating Knowledge to Resolve Uncertainty." ACM Computing Surveys 12 (2): 213–253. https://doi.org/10.1145/356810.356816.

Goldstein, Simon, and Cameron Domenico Kirk-Giannini. 2026. "A Case for AI Consciousness: Language Agents and Global Workspace Theory." Journal of Consciousness Studies 33 (7–8): 61–96. https://doi.org/10.53765/20512201.33.7.061.

Gurnee, Wes, Nicholas Sofroniew, Adam Pearce, Mateusz Piotrowski, Isaac Kauvar, Runjin Chen, Anna Soligo, Paul Bogdan, Euan Ong, Rowan Wang, T. Ben Thompson, David Abrahams, Subhash Kantamneni, Emmanuel Ameisen, Joshua Batson, and Jack Lindsey. 2026. "Verbalizable Representations Form a Global Workspace in Language Models." Transformer Circuits Thread, July 6, 2026. https://transformer-circuits.pub/2026/workspace/index.html.

Hurlburt, Russell T., and Sarah A. Akhter. 2008. "Unsymbolized Thinking." Consciousness and Cognition 17 (4): 1364–1374. https://doi.org/10.1016/j.concog.2008.03.021.

Lindsey, J. 2025. "Emergent Introspective Awareness in Large Language Models." Transformer Circuits Thread, October 29, 2025. https://transformer-circuits.pub/2025/introspection/index.html.

Nostalgebraist, 2020. "Interpreting GPT: The Logit Lens." LessWrong, August 30, 2020. https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens.

Prinz, J. 2012. The Conscious Brain: How Attention Engenders Experience. New York: Oxford University Press.

Tye, M. 1995. Ten Problems of Consciousness: A Representational Theory of the Phenomenal Mind. Cambridge, MA: MIT Press.